

AI can identify your personality based on your selfies

Russian researchers have revealed that AI can infer people's personalities from selfies better than human raters do. The study, published in the journal *Scientific Reports*, revealed that personality predictions based on female faces appeared to be more reliable than those for male faces. The technology can be used to find 'best matches' in customer service, dating or online tutoring, the researchers from HSE University and Open University in Russia, said. Studies asking human raters to make personality judgments based on photographs have produced inconsistent results.

According to the study, there are strong theoretical and evolutionary arguments to suggest that some information about personality characteristics, particularly those essential for social communication, might be conveyed by the human face.

After all, face and behaviour are both shaped by genes and

hormones, and social experiences resulting from one's appearance may affect one's personality development. However, the recent evidence from neuroscience suggests that instead of looking at specific facial features, the human brain processes images of faces in a holistic manner.

For the findings, the researchers teamed up with a Russian-British business start-up BestFitMe to train a cascade of artificial neural networks to make reliable personality judgments based on photographs of human faces. The performance of the resulting model was above that discovered in previous studies, which used machine learning or human raters. The AI was able to make above-chance judgments about conscientiousness, neuroticism, extraversion, agreeableness, and openness based on selfies the volunteers uploaded online. The resulting personality judgments were consistent across different photographs of the same individuals.

Vodafone Idea launches voice-based contactless recharge option

Vodafone Idea has launched a voice-based contactless recharge facility for its prepaid users. This new feature allows users to recharge their phones without contacting a retailer. Customers or retailers can simply mention a ten-digit mobile number on the device, and the Google voice-enabled feature will capture the command from a distance of up to three metres, according to the telecom major.

When customers come to a retailer for recharge, the retailer usually hands over the phone (with the Smart Connect Retailer App opened) to them to type in their mobile number to ensure accuracy in entering the number. However, it is not a practical option in the current times of social distancing.

"As a customer-oriented telecom operator, it is our

constant endeavour to come up with products and services relevant to the times and keep our customers connected at all times. In line with our digital-first approach, we are digitising our processes to offer convenient and efficient services to our nearly 300 million customers. The industry-first, voice-based contactless recharge enables recharges without touch and is extremely relevant in these times when maintaining social distancing is the need of the hour to stay safe," Ambrish Jain, chief operating officer, Vodafone Idea, said in a statement.

Currently, the voice-based recharge feature supports Hindi and English languages and can take commands of mobile numbers in different variations.

Bluetooth flaw allows impersonation of trusted devices

A flaw in a Bluetooth protocol is leaving millions of devices vulnerable to attacks, according to a study released by a Swiss research institute. The vulnerability, called Bluetooth Impersonation AttackS (BIAS), allows an intrusion by an attacker posing as a previously trusted Bluetooth device.

"In this paper, we demonstrate that the Bluetooth standard contains vulnerabilities enabling an attacker to impersonate a device and to establish a secure connection with a victim, without possessing the long term key shared by the impersonated device and the victim," researchers at the Swiss Federal Institute of Technology Lausanne mention in their report.

The stealth attack does not require great sophistication. Researchers say that a hacker needs little more than a Raspberry Pi to hijack a laptop, smartwatch, cellular phone, or earphones. More than 28 Bluetooth chips on nearly three dozen devices were found to be vulnerable. These include chips from Apple, Cypress, Qualcomm, Intel, Samsung, and CSR. The vulnerability was reported to manufacturers last December. Some developed workarounds immediately and

provided updates for users.

When two Bluetooth devices enter pairing mode, a persistent (long-term) encryption key is exchanged and stored. That is why smartphone users, for instance, see a list of previously established connections on their Bluetooth setup screens that permit instant connection to known sources and bypass lengthy, repetitive setup procedures.

The flaw rests with a device's failure to ensure the authenticity of a malicious device posing as a known player utilising a captured long-term encryption key. For one thing, the Bluetooth secure connection is not encrypted; in addition, mutual authentication is not required on subsequent hookups, and devices using secure connections can rely on older, less secure connection protocols that allow access to hackers. The attack focuses on the Bluetooth Classic protocol supporting basic rate and enhanced data rate modes. The Bluetooth Special Interest Group (SIG) that oversees Bluetooth protocols says it will be updating the Bluetooth core specification covering mutual authentication rules and tightening security protocols.